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Questions

Q1.

This question is about nitrogen.

(a) The table shows the successive ionisation energies of nitrogen.

Ionisation number	Ionisation energy / kJ mol^{-1}	log (ionisation energy)
1	1402	3.15
2	2856	3.46
3	4578	3.66
4	7475	3.87
5	9445	3.98
6	53 268	
7	64 362	

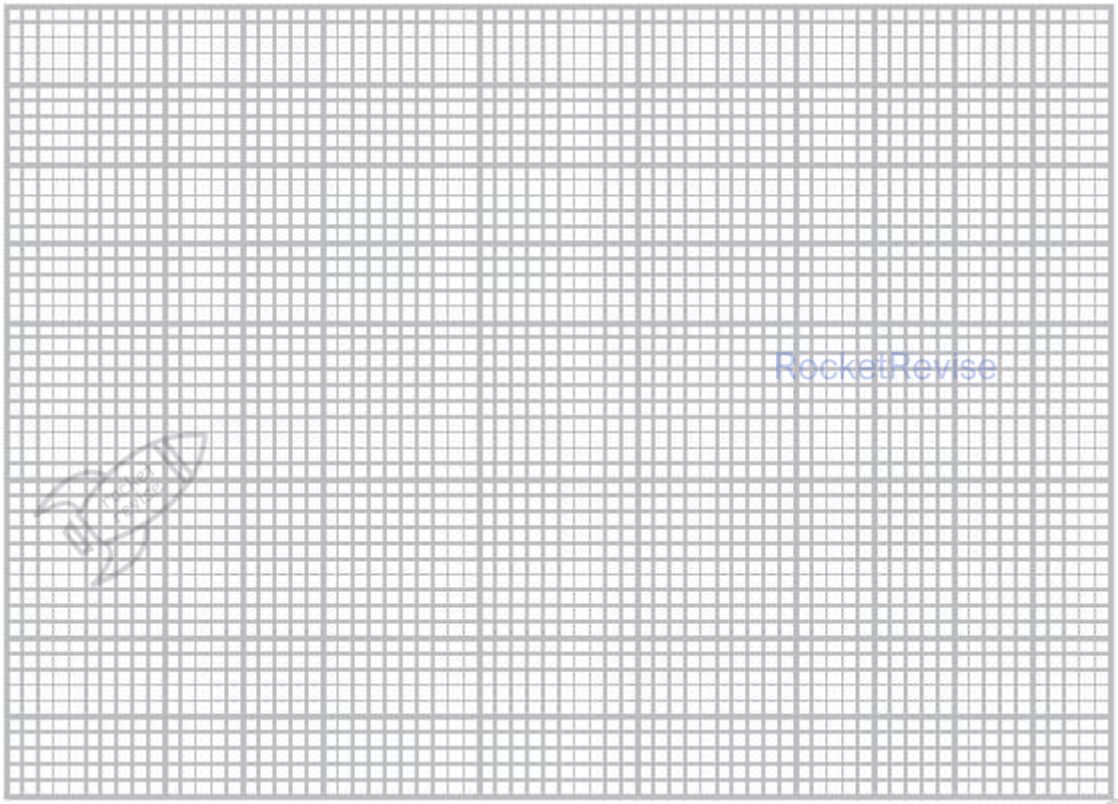
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(i) Complete the table.

(1)

(ii) Plot a graph of log (ionisation energy) against ionisation number.

(3)



(iii) Give a reason why the logarithm of the ionisation energy, rather than just the ionisation energy, is used to plot this graph.

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(iv) Explain what can be deduced from the graph about the electronic structure of nitrogen.

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(v) Explain why the first ionisation energy of oxygen is lower than that of nitrogen.

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(b) Nitrogen gas consists of nitrogen molecules.

(i) Draw a dot-and-cross diagram to show the bonding in a molecule of nitrogen.

(1)

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(ii) Calculate the number of nitrogen **atoms** in 5.60 g of nitrogen gas.
[Avogadro constant = $6.02 \times 10_{23}$ mol^{-1}]

(2)

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(iii) A sample of nitrogen gas occupied 108 cm^3 at a temperature of 25 °C and a pressure of 1.36×10^5 Pa.
Using the ideal gas equation, calculate the number of moles of nitrogen gas in this sample.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

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(Total for question = 18 marks)

Q2.

This question is about atomic structure and gases.

(a) Chlorine exists as two isotopes with mass numbers 35 and 37.

(i) State the number and type of each of the particles in the **nucleus** of a chlorine-35 atom.

(2)

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(ii) Complete the electronic configuration of a chloride **ion**, Cl^-, using the s, p, d notation.

(1)

$1s^2$ .....

(iii) A sample of chlorine contains 75.53 % of chlorine-35 atoms.
Calculate the relative atomic mass of this sample of chlorine.
Give your answer to **two** decimal places.

(2)

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(b) Fluorine has one naturally occurring isotope with mass number 19.

Chlorine and fluorine react to form chlorine trifluoride, ClF_3 .

(i) Draw a dot-and-cross diagram to show the bonding in a molecule of chlorine trifluoride.
Show the outer shell electrons only.

(2)

(ii) State, in terms of electrons, what is unusual about the dot-and-cross diagram you have drawn.

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(1)

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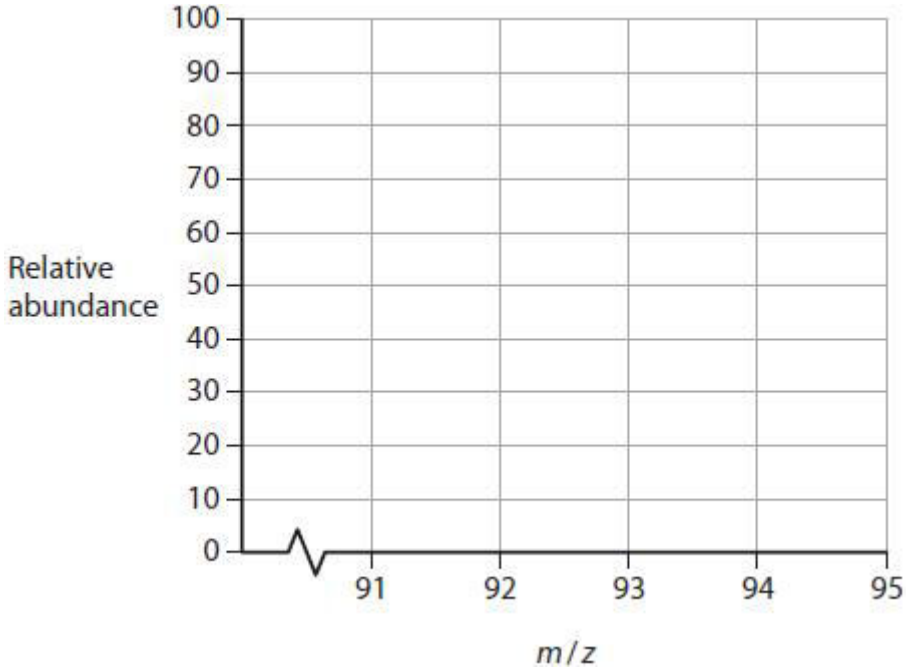
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(iii) Complete the mass spectrum to show the peaks you would expect for the molecular ion ClF_3^+ .

(3)

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(iv) Calculate, using the ideal gas equation, the volume in **cm^3** occupied by 0.0200 mol of ClF_3 gas at a temperature of 60 °C and a pressure of 1.28×10^5 Pa.

Give your answer to an appropriate number of significant figures.

[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

(Total for question = 15 marks)

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Q3.

The bonding **within** an ammonium ion, NH_4^+, is formed by

(1)

- ☒ **A** covalent bonding only
- ☒ **B** covalent and dative covalent bonding only
- ☒ **C** covalent and ionic bonding only
- ☒ **D** covalent, dative covalent and ionic bonding

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(Total for question = 1 mark)

Q4.

Which of these molecules is the **most** polar?

- ☒ **A** H—H
- ☒ **B** H—F
- ☒ **C** H—Cl
- ☒ **D** H—Br

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(Total for question = 1 mark)

Q5.

What is the polarity of the Al—Cl bond and the polarity of a trigonal planar $AlCl_3$ molecule?

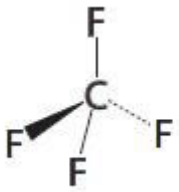
	Polarity of Al—Cl bond	Polarity of $AlCl_3$ molecule
☒ **A**	non-polar	non-polar
☒ **B**	non-polar	polar
☒ **C**	polar	non-polar
☒ **D**	polar	polar

(Total for question = 1 mark)

Q6.

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Which is correct for tetrafluoromethane (CF_4)?



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	Bonds	Molecule
☒ A	polar	polar
☒ B	non-polar	polar
☒ C	polar	non-polar
☒ D	non-polar	non-polar

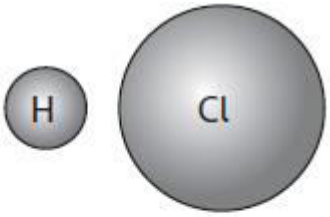
(Total for question = 1 mark)

Q7.

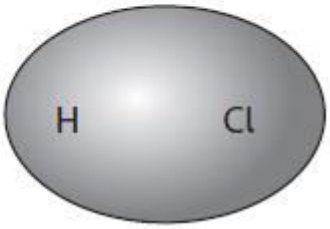
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which diagram best represents the electron density in a molecule of hydrogen chloride?

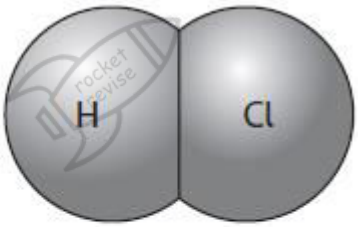
☒ A



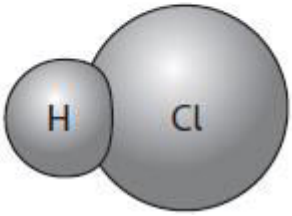
☒ B



☒ C



☒ D



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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these does **not** have a structure formed by a giant lattice of carbon atoms?

☒ **A** C_{60} fullerene

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☒ **B** diamond

☒ **C** graphene

☒ **D** graphite

(Total for question = 1 mark)

Q9.

Which row gives the correct polarities of the S—F bond and the SF_6 molecule?

	Polarity of S—F bond	Polarity of SF_6 molecule
☒ **A**	polar	polar
☒ **B**	polar	non-polar
☒ **C**	non-polar	polar
☒ **D**	non-polar	non-polar

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has bonds that are the most polar?

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- ☒ **A** H_2O
- ☒ **B** H_2S
- ☒ **C** NH_3
- ☒ **D** PH_3

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these molecules is the **most** polar?

- ☒ **A** HF
- ☒ **B** OF_2
- ☒ **C** BF_3
- ☒ **D** CF_4

(Total for question = 1 mark)

Q12.

This question is about silicon and carbon.

(a) Silicon is a semiconductor.

(i) Data obtained using the mass spectrum of silicon are shown.

Isotope mass number	Relative abundance
28	92.17
29	4.71
30	3.12

Calculate the relative atomic mass of silicon to **two** decimal places.

(2)

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(ii) Suggest a reason why there is a small peak in the mass spectrum of silicon at $m/z = 14$

(1)

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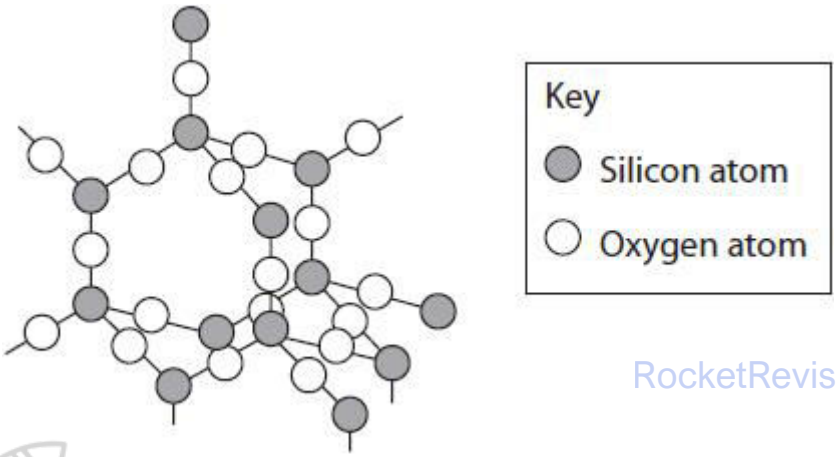
(iii) Complete the table to show the number of protons and neutrons in each isotope of silicon.

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Isotope	Number of protons	Number of neutrons
^{28}Si		
^{29}Si		
^{30}Si		

(1)

(b) Silicon dioxide, SiO_2, is the main constituent of sand and has a giant lattice structure similar to that of diamond.



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Crystalline silicon dioxide is used on the surface of semiconductor devices to provide a heat-resistant, electrically insulating layer.
Explain how the structure and bonding of silicon dioxide make it useful for this application.

(3)

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(c) Calcium silicate is formed in the removal of silicon dioxide impurities in the extraction of iron from its ores. A sample of calcium silicate composed of calcium, silicon and oxygen was found to contain 12.0 g of calcium, 8.43 g of silicon and 14.47 g of oxygen.

Determine the empirical formula of calcium silicate.
You **must** show your working.

(3)

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(d) Carbon dioxide is a gas at room temperature. A fizzy drink is canned at 5.0 °C and 1.3×10^5 Pa and contains approximately 3 g of carbon dioxide.

Calculate the volume, in cm^3, occupied by 3.00 g of carbon dioxide gas at 5.0 °C and 1.3×10^5 Pa.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

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(Total for question = 14 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ammonium iron(II) sulfate, $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$, is a double salt that is used as a source of iron(II) ions.

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(a) What is the relative formula mass of the double salt?

[A_r values: H = 1.0 N = 14.0 O = 16.0 S = 32.1 Fe = 55.8]

(1)

- ☐ **A** 277.9
- ☐ **B** 284.0
- ☐ **C** 392.0
- ☐ **D** 447.8

(b) Ammonium sulfate is used in the preparation of the double salt.

What types of bond are present in ammonium sulfate?

(1)

- ☐ **A** ionic only
- ☐ **B** covalent and ionic only
- ☐ **C** dative covalent and ionic only
- ☐ **D** ionic, covalent and dative covalent

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(c) What is the **total** number of ions present in 0.1 mol of the double salt?

[Avogadro constant (L) = 6.02×10^{23} mol^{-1}]

(1)

- ☐ **A** 1.80×10^{23}
- ☐ **B** 2.41×10^{23}
- ☐ **C** 3.01×10^{23}
- ☐ **D** 6.62×10^{23}

(Total for question = 3 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which substance has a giant lattice of atoms?

- ☐ **A** diamond
- ☐ **B** ice
- ☐ **C** poly(ethene)
- ☐ **D** sodium chloride

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(Total for question = 1 mark)

Q15.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a

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cross ☒.

Which of these molecules is **not** polar?

- ☐ **A** CO_2
- ☐ **B** HCl
- ☐ **C** H_2O
- ☐ **D** NH_3

(Total for question = 1 mark)

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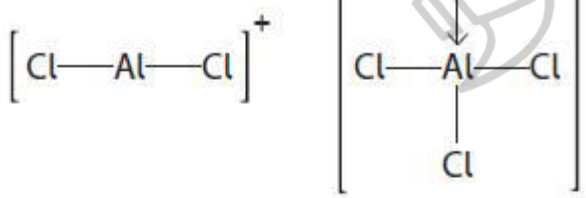
Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

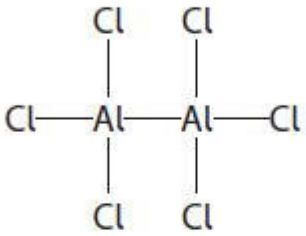
At 180°C, aluminium chloride exists as Al_2Cl_6.

What is the structure of Al_2Cl_6?

☐ **A**

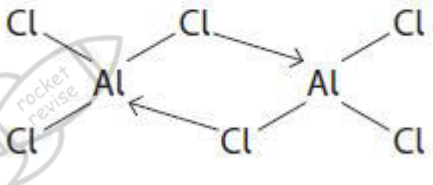


☐ **B**

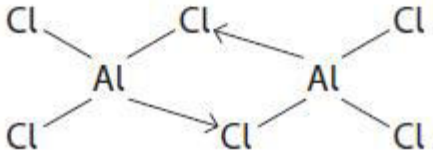


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☐ **C**



☐ **D**



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(Total for question = 1 mark)

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Q17.

Naturally occurring bromine has two isotopes: bromine-79 and bromine-81.

(a) State what is meant by the term isotopes.

(1)

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(b) Complete the table to show the numbers of subatomic particles in the two isotopes of bromine.

(2)

Isotope	Protons	Neutrons	Electrons
bromine-79			
bromine-81			

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(c) The mass spectrum of a sample of bromine is obtained.

(i) Draw a dot-and-cross diagram to show the bonding in a molecule of bromine.
Only the outer electrons should be shown.

(2)

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(ii) Describe the formation of the molecular ion of bromine in the mass spectrometer. Include an equation.
State symbols are not required.

(2)

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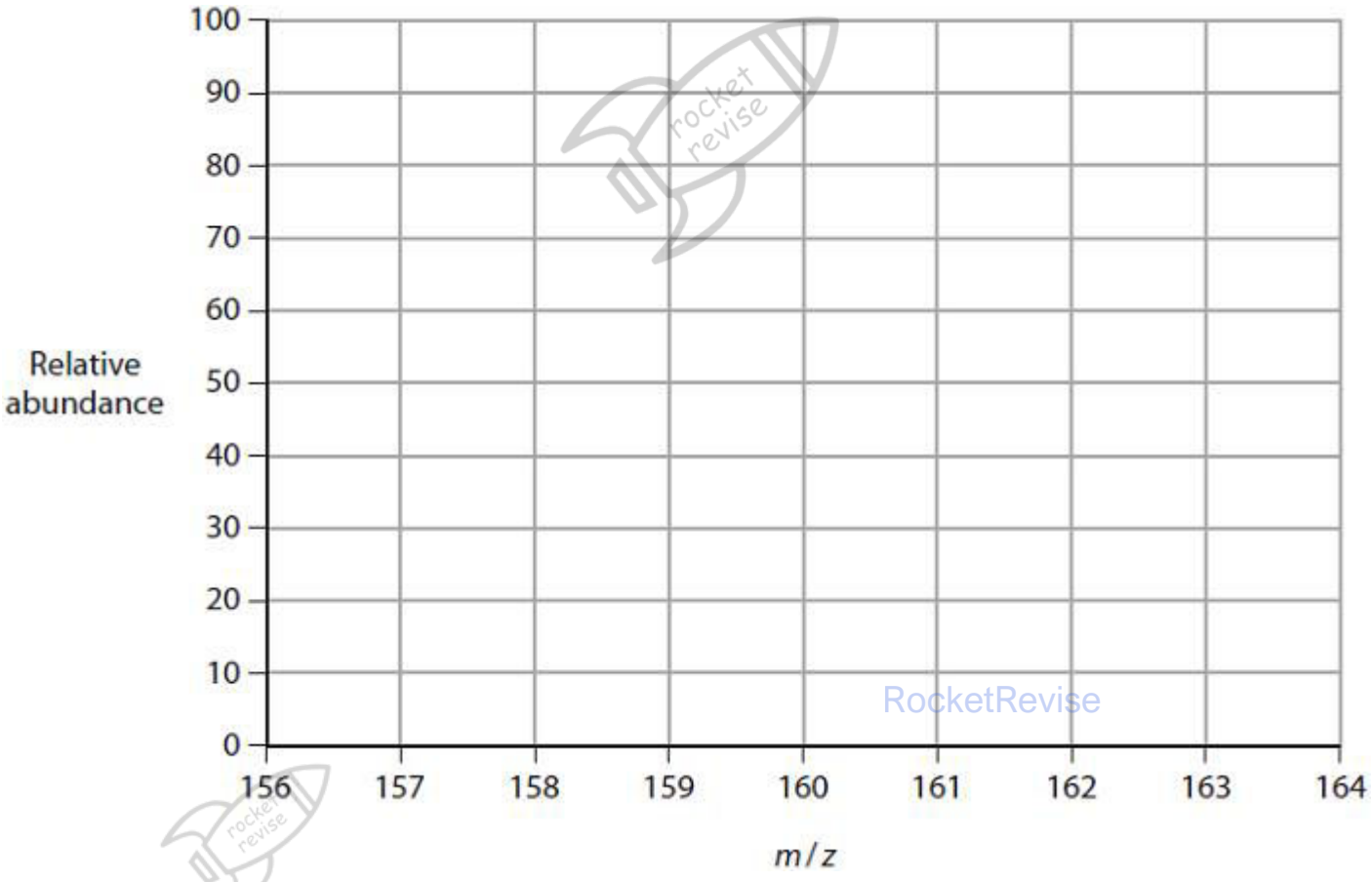
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(iii) On the mass spectrum grid, draw the peaks for the bromine molecular ions, showing the relative peak heights.
The bromine isotopes in this sample have the **same** relative abundance.

(2)



(d) The percentage abundances of the isotopes in a different sample of bromine are shown.

Isotope	Percentage abundance
bromine-79	56.38%
bromine-81	43.62%

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Calculate the relative **molecular** mass of this sample of bromine, giving your answer to **two** decimal places.

(3)

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(Total for question = 12 marks)

Q18.

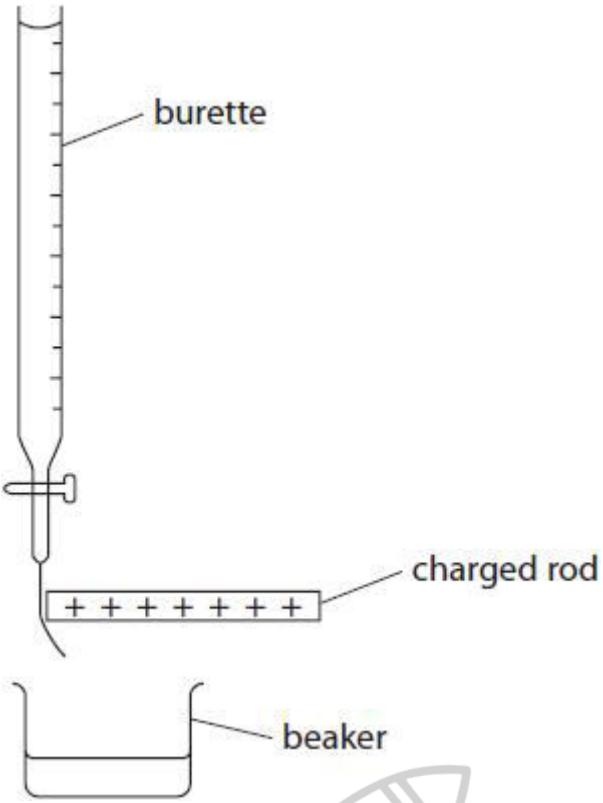
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A stream of liquid from a burette is deflected by a charged rod.

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Which substance, in its liquid state, will be deflected the most?

- ☒ **A** C_5H_{12}
- ☒ **B** CCl_4
- ☒ **C** BCl_3
- ☒ **D** CH_2Cl_2

(Total for question = 1 mark)

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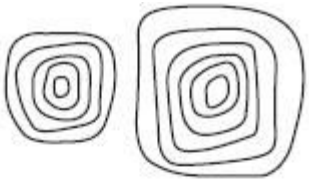
Q19.

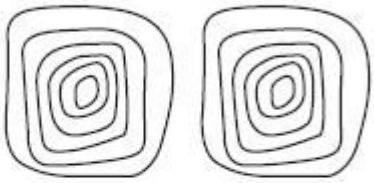
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which diagram best represents the electron density map of sodium chloride?

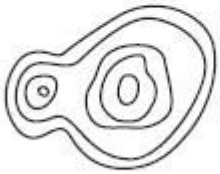
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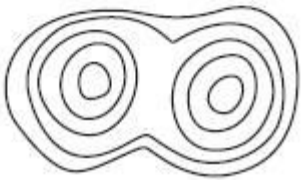
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☒ **A** 

☒ **B** 

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☒ **C** 

☒ **D** 

(Total for question = 1 mark)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule is polar?

☒ **A** CO_2
☒ **B** SF_6
☒ **C** SO_2
☒ **D** $SiCl_4$

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(Total for question = 1 mark)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule is polar?

- ☒ **A** 1,1-dibromopropane
- ☒ **B** *E*-1,2-dibromoethene
- ☒ **C** hexafluoroethane
- ☒ **D** tetrachloromethane

(Total for question = 1 mark)

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Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which property shows a **general decrease** across the Periodic Table from sodium to chlorine?

- ☒ **A** atomic radius
- ☒ **B** electronegativity
- ☒ **C** first ionisation energy
- ☒ **D** melting temperature

(Total for question = 1 mark)

Q23.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is a use of graphene based on its electrical properties?

- ☒ **A** drill tips
- ☒ **B** pencils
- ☒ **C** touch screens
- ☒ **D** body armour

(Total for question = 1 mark)

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Q24.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What types of bonding are present in the compound ammonium chloride, NH_4Cl?

	Ionic	Covalent	Dative covalent
☐ A	✓	×	✓
☐ B	×	✓	×
☐ C	✓	✓	×
☐ D	✓	✓	✓

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(Total for question = 1 mark)

Q25.

Water reacts with H^+ ions to form H_3O^+ ions.

Identify the bonding **within** the H_3O^+ ion.

(1)

- ☐ **A** covalent bonding only
- ☐ **B** covalent and dative covalent bonding only
- ☐ **C** covalent, dative covalent and ionic bonding
- ☐ **D** ionic bonding only

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(Total for question = 1 mark)

Q26.

Which describes the polarity of the C—Cl bond and the polarity of the CCl_4 molecule?

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	Polarity of C—Cl bond	Polarity of CCl_4 molecule
☐ A	non-polar	non-polar
☐ B	non-polar	polar
☐ C	polar	polar
☐ D	polar	non-polar

(Total for question = 1 mark)

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Q27.

(a) Barium chloride, $BaCl_2$, can be prepared by the reaction of barium carbonate with hydrochloric acid.

$$BaCO_3(s) + 2HCl(aq) \rightarrow BaCl_2(aq) + CO_2(g) + H_2O(l)$$

(i) Write the ionic equation for this reaction.
State symbols are not required.

(1)

(ii) Calculate the atom economy by mass for this preparation of barium chloride.
[A_r values: H = 1.0 C = 12.0 O = 16.0 Cl = 35.5 Ba = 137.3]

(2)

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(iii) Barium chloride has a melting temperature of 962 °C.
Caesium chloride has a melting temperature of 646 °C.
Explain, by considering the ions involved, the difference between the melting temperatures of these Period 6 chlorides.

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(iv) Covalent and ionic bonding are the extremes of a continuum of bonding type.
Explain the difference in bonding between barium chloride and beryllium chloride, using the electronegativity values shown.

Element	Electronegativity
Ba	0.9
Be	1.5
Cl	3.0

(2)

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(b) Barium chloride reacts with Group 1 sulfates, M_2SO_4.

$$M_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2MCl(aq)$$

A solution is made by dissolving 7.98 g of a Group 1 sulfate in deionised water.
Excess aqueous barium chloride is added to this solution and the precipitate is filtered, dried and weighed.
The mass of the barium sulfate precipitate is 10.72 g.
Identify the Group 1 element.
[A_r values: Ba = 137.3 S = 32.1 O = 16.0]

(4)

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(Total for question = 13 marks)

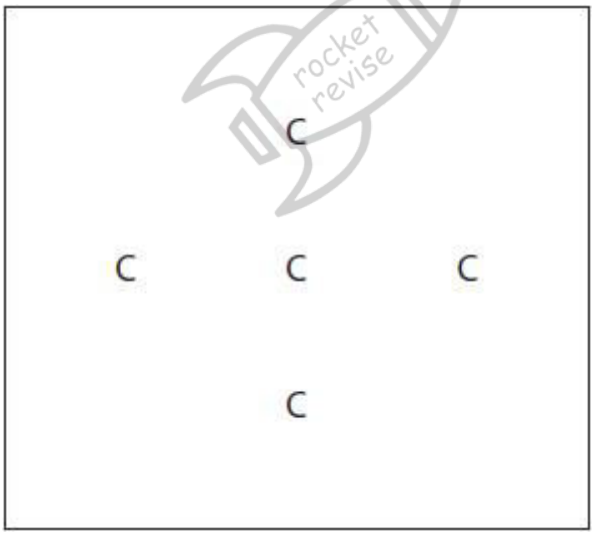
Q28.

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Carbon exists as several different structures called allotropes.

(a) Complete the dot-and-cross diagram by adding all the electrons in the outer shells of the five carbon atoms in a diamond tetrahedral unit.

(2)



(b) Draw a labelled 3D structure of graphite, showing two layers of at least 13 carbon atoms each and the forces between the layers.

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(2)

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(c) Suggest why diamond has a greater density than graphite.

(2)

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(Total for question = 6 marks)

Q29.

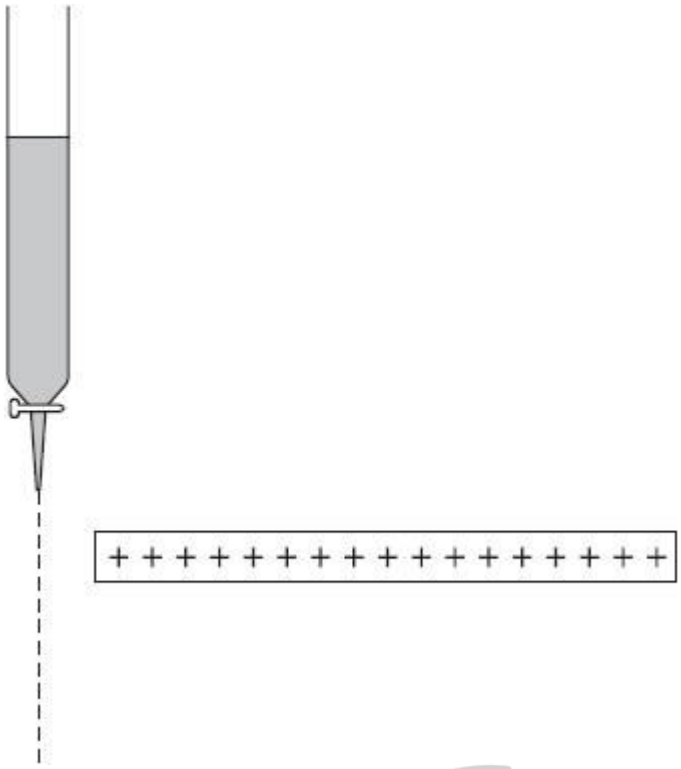
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

To compare polar and non-polar liquids, a stream of liquid from a burette and a positively charged plastic rod can be used. Which row of the table shows the correct direction of deflection?

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		Direction of deflection	
		Polar liquid	Non-polar liquid
☒	A	towards the rod	away from the rod
☒	B	away from the rod	no deflection
☒	C	towards the rod	no deflection
☒	D	no deflection	towards the rod

(Total for question = 1 mark)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these molecules are polar?

I hydrogen chloride, HCl(g)
II beryllium chloride, $BeCl_2(g)$
III boron trichloride, $BCl_3(g)$

☒ **A** I only
☒ **B** II and III
☒ **C** I and III
☒ **D** I, II and III

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(Total for question = 1 mark)

Q31.

The periods in the Periodic Table show trends in physical properties.

(a) (i) Explain the general trend in first ionisation energies for the Period 2 elements.

(2)

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(ii) Explain which **one** of the elements from **lithium** to **nitrogen** deviates from this general trend.

(3)

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(b) Identify the **Period 3** element that has the following successive ionisation energies.

Ionisation energy / kJ mol^{-1}							
First	Second	Third	Fourth	Fifth	Sixth	Seventh	Eighth
1000	2251	3361	4564	7012	8496	27 107	31 671

(1)

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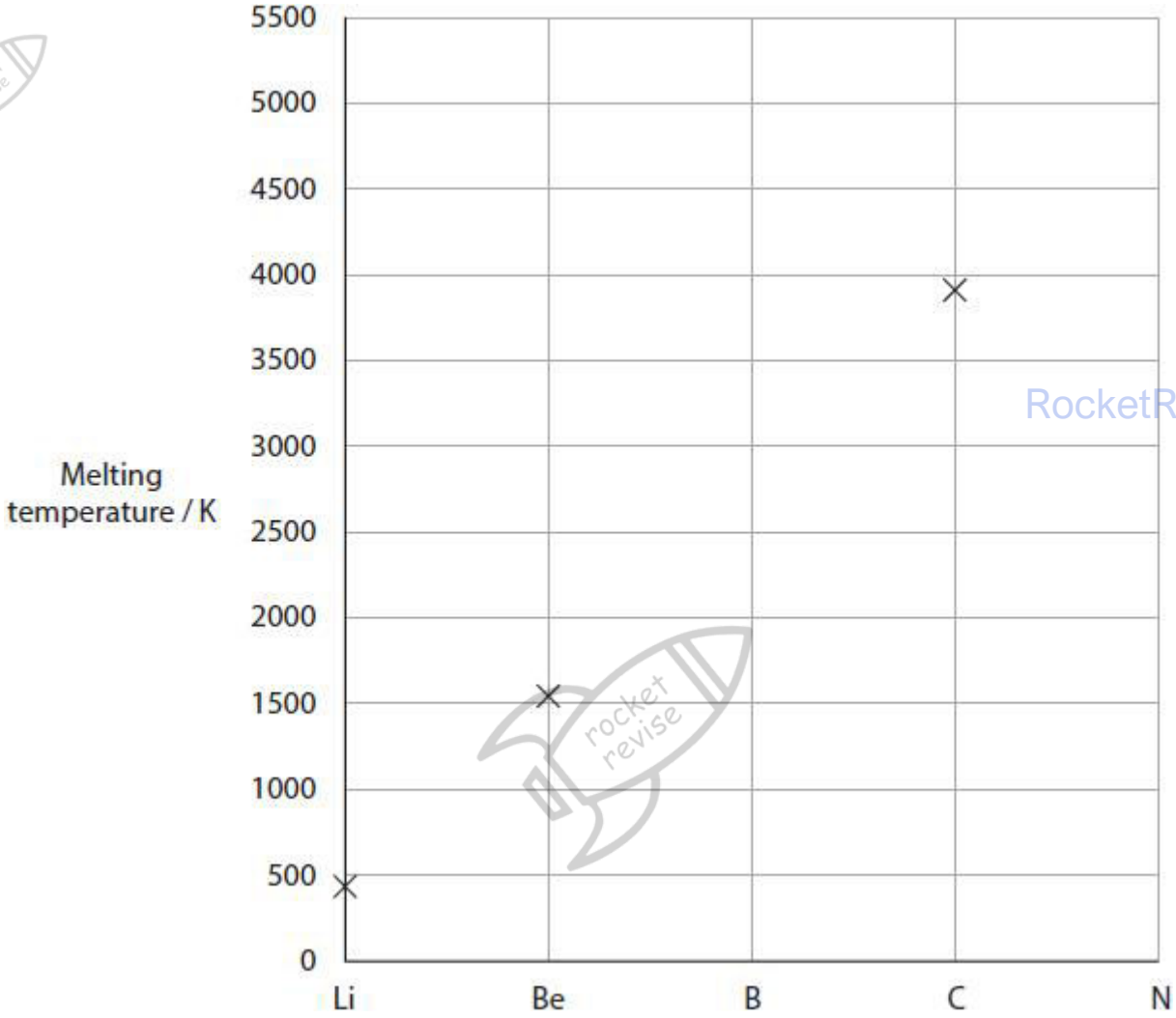
(c) (i) The diagram shows the melting temperatures of the elements in Period 2 from lithium to nitrogen.

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Complete the diagram by putting crosses to show the approximate melting temperatures of boron and nitrogen.

(2)



(ii) Explain why the melting temperature of carbon is high, with reference to its structure and bonding.

(3)

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(Total for question = 11 marks)

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Q32.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about the Period 3 elements Na, Mg, Al, Si, P, S and Cl.

(a) Which statement is **not** correct for these Period 3 elements?

(1)

- ☐ **A** atoms of Cl(g) have the highest first ionisation energy
- ☐ **B** ions of Cl^-(g) and S^{2-}(g) have the same ionic radius
- ☐ **C** atoms of Na(g) have the largest atomic radius
- ☐ **D** atoms of P(g) have the most unpaired electrons

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(b) The melting temperatures of these elements are shown in the table.

Element	Na	Mg	Al	Si	P	S	Cl
Melting temperature / K	371	922	933	1683	317	392	172

Which of the elements has the strongest **intermolecular** forces?

(1)

- ☐ **A** Al
- ☐ **B** Si
- ☐ **C** P
- ☐ **D** S

(Total for question = 2 marks)

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